

Inadvertent exposure to xenoestrogens.

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Over the last 40 years there have been constant reports concerning environmental chemicals with hormone-like effects in wildlife. An endocrine disruptor is an exogenous substance that causes adverse health effects in an intact organism or its progeny, secondary to changes in endocrine function. Endocrine disruptors of widely diverse chemical structures that have oestrogenic properties are known as oestrogenic xenobiotics or xenoestrogens. Some of these substances, such as phytoestrogens and mycoestrogens, can come from diet or from the environment. Although the oestrogenic activity of these substances is weaker than that of oestradiol, new chemicals with endocrine disrupting potential continue to be discovered, inadvertent forms of exposure are constantly being identified, and there is increasing concern about cumulative effects. Studies in the 1960s and 1970s characterized the oestrogenicity of a number of industrial compounds and the pesticides o,p-DDT, kepone, methoxychlor, phenolic derivatives and polychlorinated biphenyls (PCBs). In the last 5 years, several environmental chemicals have been added to the list of xenoestrogens, including the pesticides toxaphene, dieldrin and endosulphan, and several different compounds used in the food industry, antioxidants such as t-butylhydroxyanisole; plasticizers such as benzylbutylphthalate and 4-OH-alkylphenols; and substances used in dental restorations, such as bisphenol-A. The relevance of these newly discovered endocrine disruptors to human health is now starting to emerge. The few studies that have investigated their effect in humans point in the same direction: if there is indeed an association between exposure to substances with hormone-disruptive activity and certain disorders of endocrine organs, the incidence of such disorders would be greater in areas where exposure to agents with this activity is high. A closer scrutiny is required to determine whether these newly discovered endocrine disrupting chemicals contribute, together with oestrogenic pesticides, to the exposure of humans to xenoestrogens.